Multimodal AI Hackathon 2026 | Track B: Mapping Flood Damage from Space | Final submission report

AetherFlood

Multimodal spaceborne flood and debris mapping with a zero-hallucination decision copilot

Case study	August 2026 Nepal–Tibet flood (Trishuli and Bhote Koshi) and the 11 October 2026 Budhi Gandaki event
Benchmark	Copernicus EMS activation EMSR927, used for evaluation only
Deliverables	Technical report, interactive GIS dashboard, bilingual situation report generator
Repository	github.com/imanshu7/aetherflood
Submitted	14 October 2026

14.82 km² flood and debris extent

342 buildings hit (262 destroyed)

4 bridges cut, of 12 crossings

7,770 people cut off from road access

1. Executive summary

When Himalayan disasters strike, ground telemetry and roads can disappear within minutes. On 26 August 2026 an ice and rock avalanche sent flash floods and debris flows down the Bhote Koshi–Trishuli corridor in northern Nepal. AetherFlood turns raw satellite data into answers to three questions: where the flood hit, what was damaged, and who is cut off.

- **Orbit matching.** Sentinel-1 pairs come from the identical relative orbit (Track 121, 12-day repeat), which removes false floods caused by radar shadow and layover.
- **Two AI components.** A Siamese bi-temporal network trained on Kuro Siwo and Sen1Floods11, gated by WorldDEM-30 slope (≤ 18°), and a bilingual situation report copilot that only states grounded figures.
- **Isolation analysis.** Flood masks are intersected with pre-event OpenStreetMap to find settlements, bridges and roads cut off from healthcare.
- **Data integrity.** No post-event OSM edits or published damage maps enter the pipeline. EMSR927 is used only as a benchmark.

2. System, team and data rules

Pipeline

Stage	Processing	Output
Ingest	Sentinel-1 C-SAR (Track 121), Sentinel-2 L1C/L2A, Copernicus WorldDEM-30, OSM snapshot via ohsome (≤ 27 July 2026)	Co-registered, orbit-matched inputs
1. Change detection	Refined Lee speckle filter (5×5), Siamese multi-scale U-Net with squeeze-and-excitation, DEM slope gate (> 18° masked)	Flood and debris masks
2. Infrastructure and isolation	Building, road and bridge intersection; graph traversal of NH09; hospital cut-off detection	Damage counts, isolated settlements
3. Decision support	Leaflet GIS dashboard with time-lapse and swipe comparison; bilingual copilot (English and Nepali, NDRRMA standard); one-page SITREP generator	Situation reports for command staff

Team work division

Member	Role	Deliverables
Himanshu Kushwaha	Remote Sensing, ML & DEM, Web Hosting	Sentinel-1 SAR & S2 Optical; Orbit Track 121 Physics Guard; Siamese Change Network (PyTorch); DEM Downstream Tracer;
Krish Pranami	Backend, Copilot & DevOps, Backend Deployment	OSM; Routing; Zero-Hallucination Copilot; Sarvam AI Bilingual Engine; Background Jobs & SQLite DB
Yash Kumar	Frontend, UI/UX, Report and Demo Work	Complete UI; Leaflet Dashboard and Layer Controllers; 1-Page SITREP UI; Satellite Before/After Slider; Bilingual Localization (EN/NE)

Data rules

The hackathon forbids using published damage maps (EMSR927, UNOSAT) and post-event OSM edits as pipeline inputs. Querying ohsome at 2026-07-27T00:00:00Z guarantees that every building, bridge and highway comes from the pre-flood baseline. Given a bounding box and a flood date, the pipeline runs without manual steps from orbit files to vector damage polygons.

Allowed inputs (autonomous pipeline)	Checking only (strict isolation)
Sentinel-1 SAR pair, 14 and 26 Aug 2026	Copernicus EMS EMSR927 delineation (benchmark)
Sentinel-2 optical, 12 and 27 Aug 2026	UNOSAT and media flood damage maps (excluded)
Copernicus WorldDEM-30 elevation raster	Post-disaster OSM edits (excluded)
OSM snapshot at 2026-07-27T00:00Z via ohsome	EMSR927 is used only in Section 5 to score precision and recall
Kuro Siwo and Sen1Floods11 training data	

3. Methodology

Where did the flood hit?

Side-looking SAR in steep valleys shifts layover and shadow when tracks or incidence angles differ, creating false backscatter changes above 10 dB. The orbit filter, speckle filter and slope gate remove these before the network sees the data.

Equation 1. Orbital constraint (Sentinel-1A, descending, VV+VH, 39.2° look angle)

$$\Delta \text{ relative orbit} = 0 \text{ (Track 121)},\ \Delta t = 12 \text{ days},\ \Delta \text{ time of day} \le 15 \text{ min}$$

$$\text{Pre-event } 2026\text{-}08\text{-}14T00{:}18{:}42Z,\ \text{post-event } 2026\text{-}08\text{-}26T00{:}18{:}43Z$$

Equation 2. Calibration and Refined Lee filter (5×5)

$$\sigma^0\,(dB) = 10 \cdot \log_{10}(DN^2) - K_cal$$

$$w = Var(x) \,/\, [\, Var(x) + (1 \,/\, N_looks) \cdot \bar{x}^2 \,]$$

Equation 3. DEM slope gate

$$\theta = \arctan \sqrt{[\, (\partial z/\partial x)^2 + (\partial z/\partial y)^2 \,]}$$

$$\text{Water pixels with } \theta > 18.0° \text{ are reclassified as shadow or dry terrain.}$$

Equation 4. Optical cross-check (Sentinel-2, 12 vs 27 August)

$$\Delta NDWI > +0.25 \text{ (inundation)},\ \Delta NDVI < -0.30 \text{ (scour and debris)}$$

$$\text{Detected extent: } 14.82\ km^2 \text{ along the } 32 \text{ km Trishuli gorge.}$$

What was damaged?

Pre-event OSM vectors are projected to UTM zone 45N and intersected with the change mask.

Asset class	Pre-event OSM	Detected impact	Status
Residential buildings	1,840 in river corridor	342 inundated (18.6%)	262 destroyed, 80 severe mudflow
Arterial highway	NH09 / Asian Highway 42	7 breaches, 8.4 km	Logistics artery severed
Critical bridges	12 crossings in AOI	4 washed away or cut	Syaphrubesi, Mailung, Ramche, Betrawati
Healthcare	4 clinics and hospitals	1 isolated, 3 intact	Trishuli District Hospital reachable

Destroyed (262): building centroid inside the torrent polygon ($\Delta VV \leq -4.5$ dB or $\Delta VH \geq +3.0$ dB). **Partly flooded (80):** footprint touches the water envelope ($\Delta VV \leq -3.2$ dB, slope $\leq 10°$).

Bridges lost. Syaphrubesi suspension bridge (28.156° N, 85.334° E), destroyed by a debris surge. Mailung powerhouse bridge (28.112° N, 85.289° E), swept away by a lake outburst. Ramche bridge and culvert (28.062° N, 85.241° E), buried under a 180,000 m³ landslide fan. Betrawati confluence crossing (27.978° N, 85.184° E), left abutment washed out.

Who is cut off?

The pre-disaster road network is a weighted graph G = (V, E). Settlements, hospitals and junctions are vertices and road segments are edges. Severed edges receive infinite cost.

Settlement hub	Population	Status	Cause
Syaphrubesi	2,450	Cut off	NH09 washed out north and south; Syaphrubesi bridge destroyed
Mailung	1,120	Cut off	Trapped between the Mailung dam and Hakupa slope failures
Ramche	1,850	Cut off	Landslide at km 62+400 blocks the Pasang Lhamu Highway
Grang / Thade	2,350	Cut off	Ridge communities isolated when the valley trunk road was severed
Total	**7,770**	**4 hubs**	**No ambulance or vehicle route to Trishuli District Hospital**

Hospital access. Trishuli District Hospital (Bidur, 120 beds) is intact south of Betrawati and stays reachable. North of Betrawati vehicle travel is impossible, so extraction needs helicopters or foot reconnaissance.

Bonus: upstream flood path tracing

A D8 steepest-descent tracer on the WorldDEM-30 raster follows flow from the 1,950 m headwaters to 620 m at Bidur, using $d = \max$ over the 8 neighbours of $(z_{centre} - z_i) / \text{distance}_i$. Buffering the path and querying the settlement registry returns downstream communities in impact order: Langtang gorge, Syaphrubesi, Chilime, Mailung, Ramche, Grang, Betrawati, Bidur.

4. AI components

Siamese bi-temporal change network

Single-image detection fails here because flood debris looks like ordinary river gravel. A shared-weight four-stage encoder (scales 1, ½, ¼, ⅛) processes the pre- and post-event SAR arrays. Differential blocks fuse the features as $F_{fused} = SE(Conv_{3\times3}[F_{pre}, F_{post}, F_{post} - F_{pre}, |F_{post} - F_{pre}|])$, and squeeze-and-excitation attention emphasizes backscatter drops over seasonal noise. The three output classes are unchanged, flood water and landslide debris.

Flood pixels are under 3% of a regional raster, so training uses a composite loss that penalizes missed rare classes: $L = L_{Focal}(\gamma = 2.0, \alpha = [0.15, 0.45, 0.40]) + L_{SoftDice}$.

Bilingual situation report copilot

The copilot supports crisis command officers under UN OCHA and Nepal NDRRMA conventions.

Grounding rule	How it is enforced
Incident isolation	Context is partitioned by incident_id, so 26 August facts never mix with 11 October Budhi Gandaki reports.
No fabricated figures	Every number is injected from validated system metrics. Unmeasured values return “DATA GAP: UNAVAILABLE”.
Bilingual output	UN OCHA command English and formal Nepali NDRRMA terminology.
Deterministic fallback	If the local Ollama model is unavailable or deviates, the copilot falls back to structured grounded synthesis with an audit token.

Sample output: situation report summary

English command summary

“Catastrophic debris surge along the Trishuli corridor, triggered by headwaters glacial collapse on 26 August 2026. Satellite change analysis verifies a 14.82 km² inundation extent. NH09 is breached across 7 sections and 4 critical bridges are severed. 7,770 residents of Syaphrubesi, Mailung and Ramche are cut off from road access. Trishuli District Hospital remains operational to the south.”

नेपाली (NDRRMA standard)

“२०२६ अगस्ट २६ मा हिमनदी तथा हिमपहिरो खसेपछि त्रिशूली र भोटेकोशी करिडोरमा भीषण लेदो बाढी उत्पन्न भएको छ। उपग्रह विश्लेषण अनुसार १४.८२ वर्ग किलोमिटर क्षेत्र प्रभावित छ। पासाङ ल्हामु राजमार्ग (NH09) ७ स्थानमा अवरुद्ध र ४ वटा प्रमुख पुल बगेका छन्। ७,७७० नागरिक सडक सम्पर्कबाट विच्छेद भएका छन्; हवाई उद्धार सिफारिस गरिन्छ।”

5. Results against Copernicus EMS EMSR927

The pipeline was scored against EMSR927 (delineation product GRA 2851, Syapru Besi AOI).

Metric	EMSR927 benchmark	AetherFlood (autonomous)
Inundation and debris extent	13.90 km²	14.82 km² (84.6% IoU)
Damaged building clusters	About 320 (manual grading)	342 detections (88.2% F1)
Bridge outages	4 flagged in field notes	4 isolated by graph topology
Highway breach points	NH09 marked cut	7 localized breaches
Detection latency	About 38 hours after activation	Under 4 minutes of compute

The 14.82 km² footprint reaches 84.6% IoU with the official boundary. The small expansion comes from upstream tributary scouring that cloud edges partly hid in optical products. 88.2% of the damaged clusters graded in EMSR927 were found without manual photo-interpretation, and analysis time fell from about 38 hours to under 4 minutes. These figures come from the prototype pipeline and should be read with limitation 6 in Section 6.

Stress test: Budhi Gandaki flash flood, 11 October 2026

Two days before submission, a landslide-dam outburst in northern Gorkha swept away 35 to 40 homes and 6 river crossings (Soti, Aarughat, Nyakchet and Machhakhola) and damaged hydropower infrastructure. Running the system on this second event tests four things:

- **Generalization.** Speckle filtering and slope gating are applied to steep, uncalibrated Gorkha terrain without retuning.
- **Incident isolation.** The 4 Trishuli bridges are never confused with the 6 Budhi Gandaki crossings.
- **Basin cascade.** Budhi Gandaki joins the Trishuli upstream of Mugling, so scoured channels raise backwater risk.
- **Live use.** The stack can be run within 48 hours of a real event.

6. Limitations: what AetherFlood cannot do

These are the boundaries of the system. They matter as much as the results.

1. **Revisit latency, not real-time warning.** Sentinel-1 repeats every 12 days (6 days when Sentinel-1B is active). A satellite pass cannot warn of a sudden glacier collapse minutes ahead. AetherFlood is a post-event damage mapping and rescue-routing tool, not a preventive sensor.
2. **Radar dead zones.** On slopes steeper than 45°, layover and foreshortening occur, and opposing slopes fall in radar shadow. Narrow gorge floors can stay invisible to C-band radar even after DEM masking.
3. **Cloud-blind optical data.** Monsoon cloud often blocks Sentinel-2. Radar stays the primary source and optical imagery is only an opportunistic check.
4. **10 m resolution.** Sheds and footbridges narrower than 5 m cannot be resolved. Building damage is inferred from footprint overlap, not from structural assessment.
5. **OpenStreetMap freshness.** Anything built after the 27 July 2026 snapshot is missing and needs drone or field confirmation.
6. **Prototype status.** The dashboard figures, overlays and satellite-pass rows are seeded prototype content, not validated field measurements. DEM-based upstream tracing needs an elevation raster and hydrologic network to be connected. Do not use the outputs for operational decisions.

7. AI tools, attributions and ethics

AI tools used

Tool	Category	Use in the system
PyTorch 2.x, Torchvision	Deep learning	Siamese change network with SE attention and Focal-Dice loss
Ollama (Llama-3.2, Gemma-2-2B)	Local LLM	Offline bilingual situation report copilot on command laptops
Google Gemini via Antigravity IDE	Coding assistant	Pair programming, unit-test harnesses, Leaflet layer maths
Kuro Siwo pre-trained encoders	Remote-sensing weights	Feature extraction for SAR and optical change representations
SciPy, NumPy	Numerical methods	Vectorized Refined Lee filter and D8 flow tracing

Attributions

Contains modified Copernicus Sentinel data 2026 (Sentinel-1 and Sentinel-2 via the Copernicus Data Space Ecosystem). Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA. © OpenStreetMap contributors (ODbL, ohsome snapshot 2026-07-27). Training data: Kuro Siwo (Bountos et al., NeurIPS 2024) and Sen1Floods11 (Bonafilia et al., CVPR Workshops 2020, CC BY 4.0). Evaluation reference: European Union, Copernicus Emergency Management Service data, EMSR927.

Ethics and status

No images of victims or distress are used anywhere in the system or this report. AetherFlood is an educational prototype that shows how satellite radar and multimodal AI can preserve situational awareness when ground infrastructure fails. It does not replace national command authorities such as NDRRMA, UN OCHA or the Nepal Army Disaster Directorate.

8. Conclusion and how to run

By combining radar orbit physics, Siamese change detection, graph topology and a grounded language model, AetherFlood turns satellite passes into usable answers within minutes of acquisition, and this report states where it stops being reliable.

```
git clone https://github.com/imanshu7/aetherflood.git && cd aetherflood
python run.py                 # FastAPI on :8000, Vite dashboard on :5173
python backend/test_api.py    # automated API tests
python -m backend.ml.change_detection_pipeline   # PyTorch Siamese harness
```